1. **What do you mean by Backpropagation? What are the differences between forward and back propagation in neural network? List down the names of some popular Activation Functions used in Neural Networks.**

The backpropagation algorithm is used to train multilayer perceptrons. It propagates the error information from the end of the network to all the weights inside the network. It allows the efficient computation of the gradient or derivatives.

Backpropagation can be divided into the following steps:

It can forward the propagation of training data through the network to generate output.

It uses target value and output value to compute error derivatives by concerning the output activations.

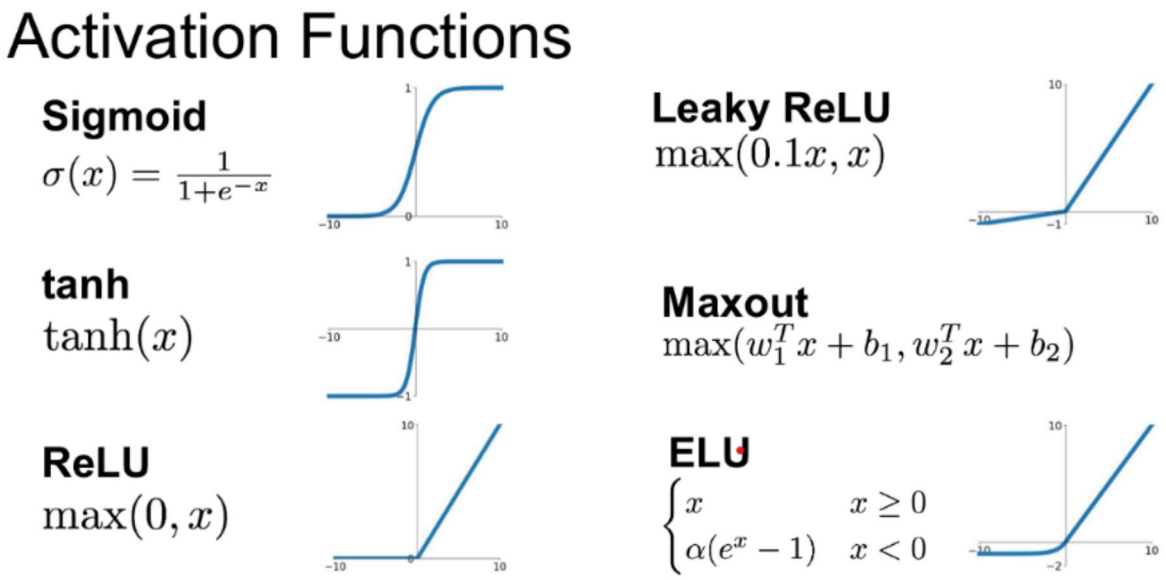
It can backpropagate to calculate the derivatives of the error concerning output activations in the previous layer and continue for all the hidden layers.

It uses the previously computed derivatives for output and all hidden layers to calculate the error derivative concerning weights.

It updates the weights and repeats until the cost function is minimized.

Forward propagation: The input is fed into the network. In each layer, there is a specific activation function and between layers, there are weights that represent the connection strength of the neurons. The input runs through the individual layers of the network, which ultimately generates an output.

Backward propagation: an error function measures how accurate the output of the network is. To improve the output, the weights have to be optimized. The backpropagation algorithm is used to determine how the individual weights have to be adjusted. The weights are adjusted during the gradient descent method.



2. **What is deep learning? What are the differences between deep learning and machine learning? What are the differences between supervised, unsupervised and reinforcement learning?**

Deep learning is a subset of machine learning, which is essentially a neural network with three or more layers. These neural networks attempt to simulate the behavior of the human brain—albeit far from matching its ability—allowing it to "learn" from large amounts of data. While a neural network with a single layer can still make approximate predictions, additional hidden layers can help to optimize and refine for accuracy.

Deep learning drives many artificial intelligence (AI) applications and services that improve automation, performing analytical and physical tasks without human intervention. Deep learning technology lies behind everyday products and services (such as digital assistants, voice-enabled TV remotes, and credit card fraud detection) as well as emerging technologies (such as self-driving cars).

1.	Machine Learning is a superset of Deep Learning	Deep Learning is a subset of Machine Learning
2.	The data represented in Machine Learning is quite different as compared to Deep Learning as it uses structured data	The data representation is used in Deep Learning is quite different as it uses neural networks(ANN).
3.	Machine Learning is an evolution of AI	Deep Learning is an evolution of Machine Learning. Basically, it is how deep is the machine learning.
4.	Machine learning consists of thousands of data points.	Big Data: Millions of data points.
5.	Outputs: Numerical Value, like classification of the score.	Anything from numerical values to free-form elements, such as free text and sound.
6.	Uses various types of automated algorithms that turn to model functions and predict future action from data.	Uses neural network that passes data through processing layers to, interpret data features and relations.
7.	Algorithms are detected by data analysts to examine specific variables in data sets.	Algorithms are largely self-depicted on data analysis once they're put into production.
8.	Machine Learning is highly used to stay in the competition and learn new things.	Deep Learning solves complex machine learning issues.
9.	Training can be performed using the CPU (Central Processing Unit).	A dedicated GPU (Graphics Processing Unit) is required for training.